

MSIM — Memory State Integrity Module

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Independent Methodological Authority

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Architecture Ecosystem: Structured Reality Standards™

Architecture Family: Memory Governance Intelligence Architecture (MGIA™)

Operational Layer: Memory Integrity Governance Layer

Governed Space: Memory State Integrity

Category: AI & Interpretation

Subcategory: Memory State Governance

Type: Memory Integrity Module

Parent Standard: Memory Integrity Standard (MIS)

Version: 1.0

Status: Canonical · Open Module

Effective Date: 8 June 2026

Compatibility: OOF Methodology OS · Memory Integrity Standard (MIS) ·
Memory Continuity Integrity Module (MCIM) · Memory Traceability Integrity Module (MTIM) ·
Memory Boundary Integrity Module (MBIM) · Temporal Memory Integrity Standard (TMIS) ·
Memory Evolution Integrity Standard (MEIS) · INTEGROS® — Integrity Standard

AI-Readable: Yes

Authority: OOF

Protection: MIP — Methodological Intellectual Property

Canonical Language: English (UCL)

Canonical Definition System

Canonical Definition

Memory State Integrity Module (MSIM) defines the structural conditions under which memory states remain coherent, identifiable, assessable, reconstructable, reality-connected, and operationally valid throughout memory creation, storage, retrieval, modification, preservation, and utilization processes.

MSIM governs memory states.

The module ensures that memory remains in a known, governable, and assessable condition rather than degrading into ambiguous, corrupted, inconsistent, unstable, or operationally invalid states.

A system satisfies MSIM only if:

- memory states remain identifiable
- memory-state transitions remain traceable
- state consistency remains assessable
- state degradation remains detectable
- memory coherence remains preservable
- memory states remain operationally valid

A memory environment that cannot determine the current state of memory does not satisfy MSIM.

Module Operational Role

MSIM defines the memory-state governance layer of MIS.

Its role is to preserve valid operational conditions of memory.

Module Operational Space

MSIM governs:

- memory states
- state transitions
- memory consistency
- memory coherence
- state degradation
- memory stability
- memory-state accountability
- state governance

The module applies wherever memory changes condition through time.

Module Function

The module applies wherever systems must preserve:

- coherent memory states
- traceable state transitions
- governance-valid memory conditions
- accountable memory modifications
- operational memory stability
- reliable memory utilization

Its function is to ensure that memory remains in a valid and governable operational state.

Minimum Implementation Framework

1. Define the Memory State Object

The organization must define which memory states require governance.

This may include:

- active memory
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- archived memory
- temporary memory
- long-term memory
- shared memory
- restricted memory
- evolving memory
- operational memory

2. Define Memory State Conditions

The system must define the conditions under which memory states remain valid.

This includes:

- state-identification requirements
- consistency requirements
- coherence requirements
- transition requirements
- stability requirements
- governance-valid state conditions

3. Define State Degradation Detection Logic

The system must define how memory-state degradation is identified.

This may include:

- memory corruption
- state inconsistency
- invalid transitions
- coherence failures
- unstable memory conditions
- governance-blind state changes

4. Define Operational Response or Governance Logic

The system must define governance logic for state-integrity failures.

Governance response may include:

- state review
- consistency verification
- memory restoration
- governance intervention
- escalation
- state correction
- operational invalidation where required

5. Preserve Traceability & Restrict Invalid Conditions

The system must preserve reconstructable traceability of:

- memory states
- state transitions
- modification events
- governance reviews
- intervention actions
- resulting memory conditions

A memory environment must not remain state-valid if materially significant memory states cannot be identified, reconstructed, reviewed, or governed.

Use Case 1 — Long-Term AI Memory Platform

Scenario

An AI platform manages active, archived, temporary, and evolving memory objects across multiple operational contexts.

Application

MSIM governs memory-state visibility, transition integrity, and operational stability across the memory environment.

Result

The platform gains stronger memory reliability, improved state transparency, and reduced exposure to memory corruption or state inconsistency.

Use Case 2 — Enterprise Knowledge Management System

Scenario

A large organization continuously creates, updates, archives, restores, and utilizes knowledge assets throughout operational activities.

Application

MSIM governs memory-state management, transition accountability, and preservation of knowledge consistency.

Result

The organization gains stronger operational continuity, improved knowledge reliability, and reduced exposure to unmanaged state degradation.

Canonical Closing Statement

Memory State Integrity Module (MSIM) defines the structural conditions under which memory states remain coherent, identifiable, assessable, reconstructable, reality-connected, and operationally valid throughout memory creation, storage, retrieval, modification, preservation, and utilization processes.

Memory cannot remain governable if its state becomes unknown. Memory state integrity therefore becomes a foundational integrity condition of governable memory systems.

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Canonical Source

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